

J. Jake Nichol, M.B.A., Ph.D.

✉ jjakenichol@gmail.com

🌐 <http://jjakenichol.com>

🔗 [jjakenichol](#)

🆔 0000-0001-9713-1469

🌐 [jjakenichol](#)



Employment History

- 2025 – ··· **Postdoctoral Appointee**, Scientific Machine Learning, Sandia National Laboratories.
- 2019 – 2025 **R&D Graduate Intern**, Scientific Machine Learning, Sandia National Laboratories.
- 2018 – 2019 **Graduate Research Assistant**, Tapia Lab, Department of Computer Science, School of Engineering, University of New Mexico.
- 2017 – 2019 **Owner**, Swarming Technologies LLC: Robotics design, manufacturing, and repair.
- 2015 – 2017 **Robot Engineer/Designer**, NASA Swarmathon & Moses Biological Computation Lab, Department of Computer Science, School of Engineering, University of New Mexico.
- 2014 – 2014 **Software Engineering Intern**, Intel Corporation.

Education

- 2018 – 2025 **Ph.D. Computer Science, University of New Mexico**, School of Engineering. Moses Biological Computation Lab.
Thesis title: *Seeking Structure in Complex Systems: From Feature Analysis to Space-Time Causal Discovery with Earth Science Applications*
- 2015 – 2017 **M.B.A., University of New Mexico**, Anderson School of Management.
- 2011 – 2016 **B.S. Computer Science, University of New Mexico**, School of Engineering.

Research Publications

Journal and Conference Articles

- 1 **J. J. Nichol**, M. Weylandt, G. M. Fricke, M. E. Moses, D. Bull, and L. P. Swiler, “Space-Time Causal Discovery in Earth System Science: A Local Stencil Learning Approach,” *Journal of Geophysical Research: Machine Learning and Computation*, vol. 2, no. 3, 2025, ISSN: 2993-5210. [DOI](#): 10.1029/2024jh000546.
- 2 **J. J. Nichol**, M. Peterson, G. M. Fricke, and K. Peterson, “Learning Why: Data-Driven Causal Evaluations of Climate Models,” *ICML 2021 Workshop Tackling Climate Change with Machine Learning*, 2021. [DOI](#): 10.2172/1884401.
- 3 **J. J. Nichol**, M. G. Peterson, K. J. Peterson, G. M. Fricke, and M. E. Moses, “Machine learning feature analysis illuminates disparity between E3SM climate models and observed climate change,” *Journal of Computational and Applied Mathematics*, vol. 395, p. 113 451, Oct. 2021, ISSN: 0377-0427. [DOI](#): 10.1016/j.cam.2021.113451.

In Preparation

- 1 **J. J. Nichol**, M. Weylandt, G. M. Fricke, A. Mueen, E. M. Moses, and P. L. Swiler, “M-CaStLe: Uncovering Local Causal Structures in Multivariate Space-Time Gridded Data.”

Technical Reports

- 1 **J. J. Nichol**, M. Weylandt, M. Smith, and L. Swiler, “Benchmarking the PCMCi Causal Discovery Algorithm for Spatiotemporal Systems,” Tech. Rep., 2023. [URL](#): <https://www.osti.gov/biblio/1991387>.

- 2 J. J. **Nichol**, M. Peterson, and K. Peterson, "Causal Evaluations for Identifying Differences between Observations and Earth System Models," Tech. Rep., 2021.  DOI: 10.2172/1820528.
- 3 K. J. Peterson, A. J. Powell, I. K. Tezaur, E. L. Roesler, J. **Nichol**, M. G. Peterson, W. L. Davis, J. D. Jakeman, D. J. Stracuzzi, and D. L. Bull, "Arctic tipping points triggering global change (ldrd final report)," Tech. Rep., Sep. 2020.  DOI: 10.2172/1669210.

Other Works

- 1 K. Peterson, W. Davis, M. Peterson, J. J. **Nichol**, K. Chowdhary, and M. D'Elia, *Advancing Sea Ice Predictability in E3SM with Machine Learning*, Apr. 2021.  DOI: 10.2172/1769655.
- 2 A. Powell, E. C. S. Acquesta, W. L. Davis, J. J. **Nichol**, I. Tezaur, K. Peterson, S. Rempe, and J. G. Huerta, *Water Cycle-Driven Infectious Diseases as Multiscale, Reliable, Continuously Updating Water Cycle Sensors*, Feb. 2021.  DOI: 10.2172/1769797.
- 3 S. M. Ackerman, G. M. Fricke, J. P. Hecker, K. M. Hamed, S. R. Fowler, A. D. Griego, J. C. Jones, J. J. **Nichol**, K. W. Leucht, and M. E. Moses, *The Swarmathon: An Autonomous Swarm Robotics Competition*, 2018.  DOI: 10.48550/arxiv.1805.08320. eprint: 1805.08320.

Miscellaneous Experience

Public Speaking Engagements

- 2025  18th U.S. National Congress on Computational Mechanics (USNCCM18). *Advancing Climate Causal Discovery: Local Space-Time Stencil Learning for High-Dimensional Data*. Chicago, Illinois, United States.
- 2022  European Seminar on Computing (ESCO). *Global Multivariate Causal Discovery for the Analysis of Emergent Properties in Earth System Models*. Pilsen, Czech Republic.
- 2021  Chesapeake Large-Scale Analytics Conference (CLSAC). *Causal Discovery for Climate Science and the Energy Exascale Earth System Model*. Virtual.
- 2020  European Seminar on Computing (ESCO). *Machine Learning to Compare Arctic Simulations with Observed Data*. Pilsen, Czech Republic.

Poster Presentations

- 2024  "Recovering the Spatial Evolution of Volcanic Aerosols." **Nichol, J. J.** (2024, May 7). Sandia National Laboratories and Los Alamos National Laboratories Annual Climate Summit. Los Alamos, New Mexico, United States.
-  "Recovering the Spatial Evolution of Volcanic Aerosols." **Nichol, J. J.** (2024, April 22). Sandia National Laboratories Analytics for Climate and Earth Sciences (ACES) Symposium. Albuquerque, New Mexico, United States.
- 2021  "Learning Why: Data-Driven Causal Evaluations of Climate Models." **Nichol, J. J.**, Peterson, M. G., Fricke, G. M., & Peterson, K. J. (2021, July 18-24). International Conference on Machine Learning (ICML). Virtual.
- 2020  "Comparing Simulated and Observed Data with Random Forest Feature Importance." **Nichol, J. J.**, Peterson, M., Peterson, K., & Stracuzzi, D. (2020, February 25-27). Conference on Data Analysis (CoDA). Santa Fe, New Mexico, United States.
- 2019  "Feature Comparison of Arctic Observations and Climate Models." **Nichol, J. J.**, Peterson, M., Peterson, K., Stracuzzi, D., Fricke, M., & Moses, M. (2019, December 9-13). American Geophysical Union (AGU) Fall Meeting. San Francisco, California, United States.

Professional Memberships

- 2023-2025  Society for Causal Inference
2019-2025  American Geophysical Union

Honors and Awards

- 2020  **Best Talk Prize**, European Seminar on Computing (ESCO)
 **3rd Place Poster Prize**, Department of Energy Conference on Data Analysis (CoDA)
2011  **Eagle Scout**, Boy Scouts of America

Professional Service

Reviewer for *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*.

Software Packages

Published

- clif: CLImate Fingerprinting. *A python library that calculates empirical orthogonal functions for mainly climate data.* Developed with Kenny Chowdhary. Available on  <https://github.com/sandialabs/clif>
- CaStLe: Causal Space-Time Stencil Learning. *Local causal space-time structures from observational data.* Available on  <https://github.com/jjakenichol/CaStLe>.

Research Interests and Related Skills

Causal Inference	Algorithmic causal inference, i.e., causal structure learning, causal discovery, or causal network learning; causal inference techniques for advancing scientific machine learning.
Machine Learning	Scientific ML; domain/physics-informed ML; ML feature importance, such as random forests Gini importance, permutation importance, drop-column importance, SHAP.
Artificial Intelligence	AI for Earth systems science; trusted AI, explainable AI, and fairness and ethics in AI.
Programming	Python and data science libraries such as NumPy, SciPy, Pandas, Xarray, DASK, and Tigramite.
Misc. Tech. Skills	L ^A T _E X, high performance computing frameworks Slurm and PBS, GNU Parallel, and MATLAB. Minor experience with Docker and Anisble.

References

Available on request